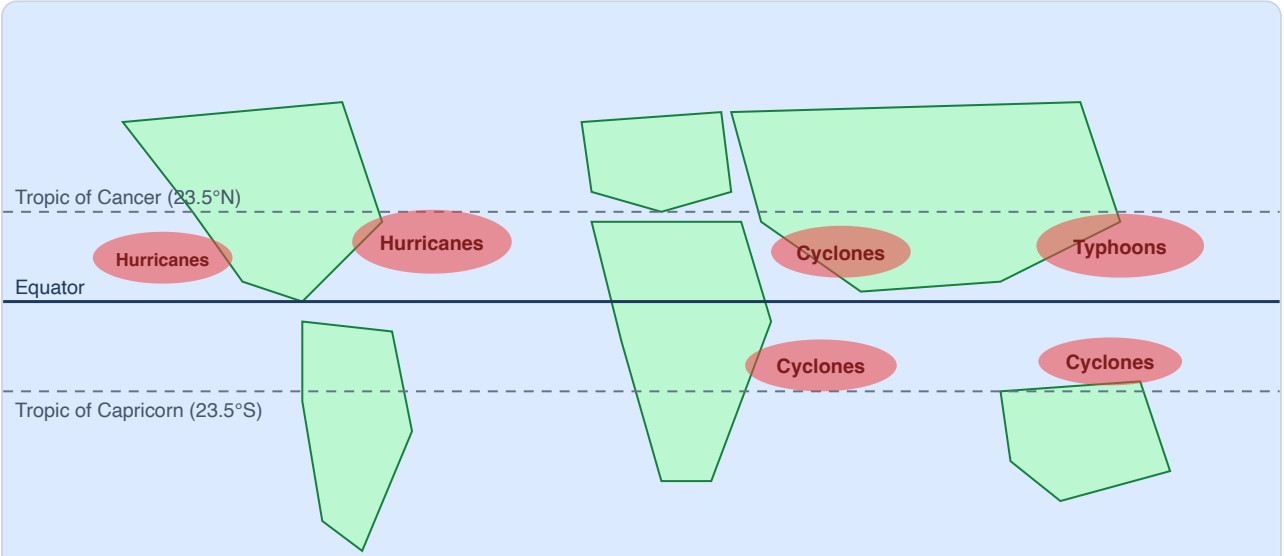
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Edexcel GCSE Geography B (1GB0) — Component 1: Global Geographical Issues · Topic 1: Hazardous Earth

1.2 Tropical Cyclones

1. Global Distribution of Tropical Storms

Tropical storm — an intense low-pressure weather system with sustained wind speeds above 119 km/h (to count as a hurricane, typhoon or cyclone), heavy rainfall and a calm central eye. They are called **hurricanes** in the North Atlantic and NE Pacific, **typhoons** in the NW Pacific, and **cyclones** in the Indian Ocean and South Pacific.



Tropical storms form between about 5° and 30° north and south of the Equator (simplified map, not to scale)

- Tropical storms form over **warm tropical oceans**, mostly between **5° and 30° north and south** of the Equator.
- They do not form **within about 5° of the Equator** because the Coriolis effect is too weak there to make the air spin.
- They move **westwards** with the trade winds, then often curve polewards. They **weaken over land** (no supply of warm, moist air) and over cooler water.

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- Most form in **late summer and autumn** (June–November in the Northern Hemisphere), when sea temperatures are highest.

2. Causes and Formation of Tropical Storms

Conditions needed

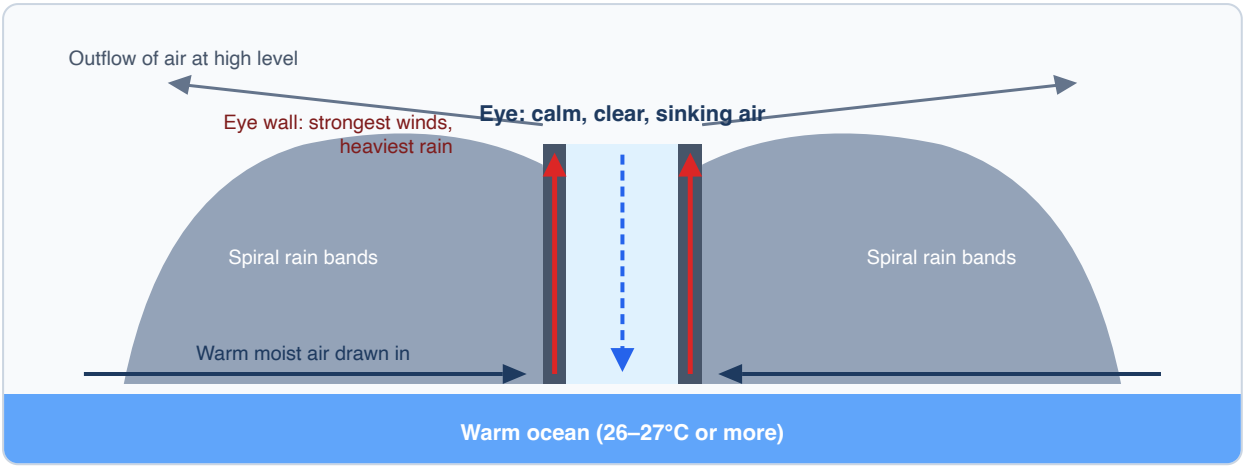
- Ocean surface temperature of **at least 26–27°C**, to a depth of about 50–60 m.
- Ocean **at least 5° from the Equator**, so the Coriolis effect can make the storm spin.
- **Low wind shear** — winds at different heights blowing at similar speeds, so the storm's structure is not torn apart.
- An area of **low pressure** with rising, unstable, moist air.

Sequence of formation

1. Warm ocean water heats the air above it. The **warm, moist air rises rapidly**, creating an area of very low pressure at the surface.
2. As the air rises it **cools and condenses**, forming towering cumulonimbus clouds. Condensation releases **latent heat**, which warms the air further and makes it rise even faster — powering the storm.
3. Air rushes in at the surface to replace the rising air, creating strong winds. The **Coriolis effect** makes the inflowing air spiral (anticlockwise in the Northern Hemisphere).
4. Some cool, dry air sinks down the centre of the storm, creating the calm, cloud-free **eye**.
5. The storm is carried across the ocean by prevailing winds, growing as long as it has a supply of warm, moist air. It weakens when it reaches **land or cooler water** because the energy supply is cut off.

Structure of a tropical storm

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Feature	Description
Eye	Centre of the storm, 10–50 km across. Calm, clear skies, lowest pressure, caused by sinking air.
Eye wall	Ring of towering cumulonimbus cloud around the eye — the most intense winds (often over 200 km/h) and heaviest rainfall.
Rain bands	Spiralling bands of cloud and rain extending hundreds of kilometres from the centre.
Size	Typically 200–700 km across; can last for 1–3 weeks.

3. Physical Hazards of Tropical Cyclones

Hazard	Effect
High winds	Sustained winds over 119 km/h (over 250 km/h in the strongest storms) destroy buildings, uproot trees and turn debris into missiles.
Intense rainfall	Hundreds of millimetres of rain in a day or two causes river and flash flooding.
Storm surges	Low pressure and onshore winds raise sea level by several metres — usually the most deadly hazard.

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Coastal flooding	Surges combined with high tides and waves flood low-lying coasts, contaminating land and water with salt.
Landslides	Saturated slopes collapse, burying homes and blocking roads.

Cyclone strength is measured on the **Saffir–Simpson scale** from category 1 (119–153 km/h) to category 5 (252 km/h or more).

4. Why Some Countries Are More Vulnerable

- **Physical factors:** low-lying coastal plains and deltas (e.g. Bangladesh, the Mississippi delta); shallow seas that increase surge heights; frequency of storms along a cyclone track.
- **Social factors:** population density, poverty, quality of housing, education and awareness of warnings.
- **Economic factors:** money for prediction, warnings, defences, shelters and recovery; insurance.
- **Political factors:** how well governments plan and respond — even wealthy countries can respond poorly.

5. How Climate Change Might Affect Tropical Storms

Aspect	Likely change	Reason
Distribution	Storms may form further from the tropics (further north and south)	A larger area of ocean will reach 26–27°C
Frequency	Total number may stay the same or fall slightly	Scientists are uncertain — changes in wind shear may suppress some storms
Intensity	A higher proportion of storms reaching the strongest categories (4 and 5), with heavier rainfall	Warmer oceans supply more energy; warmer air holds more moisture

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Storm surges	More damaging coastal flooding	Sea level rise means surges start from a higher base

Exam Tip: Be careful with certainty. The evidence for *more intense* storms is much stronger than the evidence for *more frequent* storms — say "likely" or "scientists predict" rather than "will".

6. Case Study: Typhoon Haiyan, the Philippines, November 2013

Typhoon Haiyan (known locally as Yolanda) — 8 November 2013

Background: The Philippines is a lower-middle-income country made up of over 7,000 islands in the western Pacific — one of the most typhoon-exposed countries in the world. Haiyan was one of the strongest tropical storms ever to make landfall, with sustained winds of about **315 km/h**. It hit the islands of Samar and Leyte, including the city of **Tacloban**.

Primary effects

- A **storm surge of up to 5–6 m** swept through Tacloban, flooding the city and its airport terminal.
- Over **6,300 people killed**; around 1,000 more missing; nearly 30,000 injured.
- About **1.1 million homes damaged or destroyed** — around 90% of Tacloban was destroyed.
- Around 4 million people displaced; power lines, roads and bridges destroyed.
- About 400 mm of rain caused widespread flooding.

Secondary effects

- About 14 million people affected; around 6 million workers lost their source of income.
- Flooding caused **landslides** and blocked roads, cutting off aid to remote areas.
- Shortages of food, water and shelter led to **outbreaks of disease** and some looting.
- Rice harvests and fishing fleets destroyed, causing food shortages; coconut plantations (a major source of income) flattened.
- An oil barge ran aground causing an oil spill, damaging mangroves.

Immediate responses

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- The government evacuated around **800,000 people** before landfall, using warnings from the national weather agency (PAGASA).
- International aid and teams arrived — the UK sent military aircraft and HMS Daring; the USA sent the aircraft carrier USS George Washington to supply water and food.
- Field hospitals set up; emergency shelters, food, water and hygiene kits distributed by NGOs such as the Red Cross and Oxfam.
- A curfew was imposed two days after the storm to reduce looting.

Long-term responses

- A **"Build Back Better"** reconstruction plan — rebuilding roads, bridges and airports to a higher standard.
- "Cash for work" programmes paid local people to clear debris and rebuild, restoring incomes.
- Rice farmers and fishermen given help to restart — seeds, tools and replacement boats.
- A **"no-build zone"** proposed along vulnerable coasts, with communities relocated inland.
- More **cyclone shelters** built; storm-surge warnings improved so people understood the risk (many did not know what a "storm surge" meant in 2013).
- Mangroves replanted along the coast to act as a natural barrier against future surges.

7. Case Study: Hurricane Katrina, USA, August 2005 (developed country)

Hurricane Katrina — 29 August 2005

Katrina made landfall on the US Gulf Coast as a category 3 hurricane with winds of about 200 km/h. Much of **New Orleans** lies below sea level, protected by levees, between Lake Pontchartrain and the Mississippi River.

Effects

- A storm surge of up to 8 m hit the Mississippi coast; levees in New Orleans failed in over 50 places, flooding about **80% of the city**.
- Around **1,800 people died**; over 1 million were displaced, many permanently.

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- Economic damage of about **$125 billion** — one of the costliest disasters in US history. Oil platforms and refineries in the Gulf were shut, raising fuel prices.
- The poorest residents — often without cars — were most affected; tens of thousands sheltered in the Superdome in poor conditions.

Preparation and response

- The National Hurricane Center gave accurate forecasts days in advance; a mandatory evacuation was ordered and about 80% of the population left.
- However, the federal response (FEMA) was widely criticised as slow and poorly coordinated; it took days to evacuate those left behind.
- Afterwards, the US Army Corps of Engineers spent around **$14.5 billion** on a stronger flood defence system for New Orleans, including the world's largest drainage pump station and a 3 km surge barrier.

Comparing Katrina (developed) and Haiyan (developing)

Factor	Hurricane Katrina, USA 2005	Typhoon Haiyan, Philippines 2013
Strength at landfall	Category 3	Category 5 equivalent (~315 km/h)
Deaths	About 1,800	Over 6,300
Economic cost	About $125 billion	Estimated $2–6 billion in damage (very little insured)
Key reason for impact	Levee failure and slow government response; poorer residents unable to evacuate	Huge storm surge; weak housing; poverty; poor understanding of "storm surge"

Exam Tip: Edexcel asks you to compare two cyclones in countries at different levels of development. Developed countries usually have *higher economic losses* but *fewer deaths*;

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developing countries usually the reverse. Katrina shows that wealth alone does not guarantee an effective response.

8. Reducing the Effects of Tropical Storms

Strategy	How it reduces the effects	Example
Monitoring	Satellites, aircraft and weather buoys track the storm's position, wind speed and pressure in real time	US National Hurricane Center tracks every Atlantic hurricane
Prediction	Computer models forecast the storm's path and strength several days ahead, allowing warnings and evacuation	PAGASA warnings allowed 800,000 to evacuate before Haiyan
Protection	Storm-surge barriers, sea walls, raised buildings on stilts, reinforced roofs, cyclone shelters	Bangladesh has built thousands of raised cyclone shelters, cutting death tolls dramatically since 1991
Planning	Evacuation routes, emergency supplies, education and drills; land-use zoning to keep homes away from the coast	Florida's signed hurricane evacuation routes; Philippines' no-build zones

Exam Tip: Planning and prediction save the most lives at the lowest cost — Bangladesh's 1970 Bhola cyclone killed around 300,000–500,000 people, whereas Cyclone Sidr (2007), of similar strength, killed about 3,400 thanks to early warnings and shelters. Use this contrast to evaluate strategies.

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